

CYP3A4 DDI Qualification

Version	main-OSP12.3
Qualification Plan Release	https://github.com/Open-Systems-Pharmacology/Qualification-DDI-CYP3A4/releases/tag/vmain
OSP Version	12.3
Qualification Framework Version	3.7

This qualification report is filed at:

<https://github.com/Open-Systems-Pharmacology/OSP-Qualification-Reports>

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1 Qualification

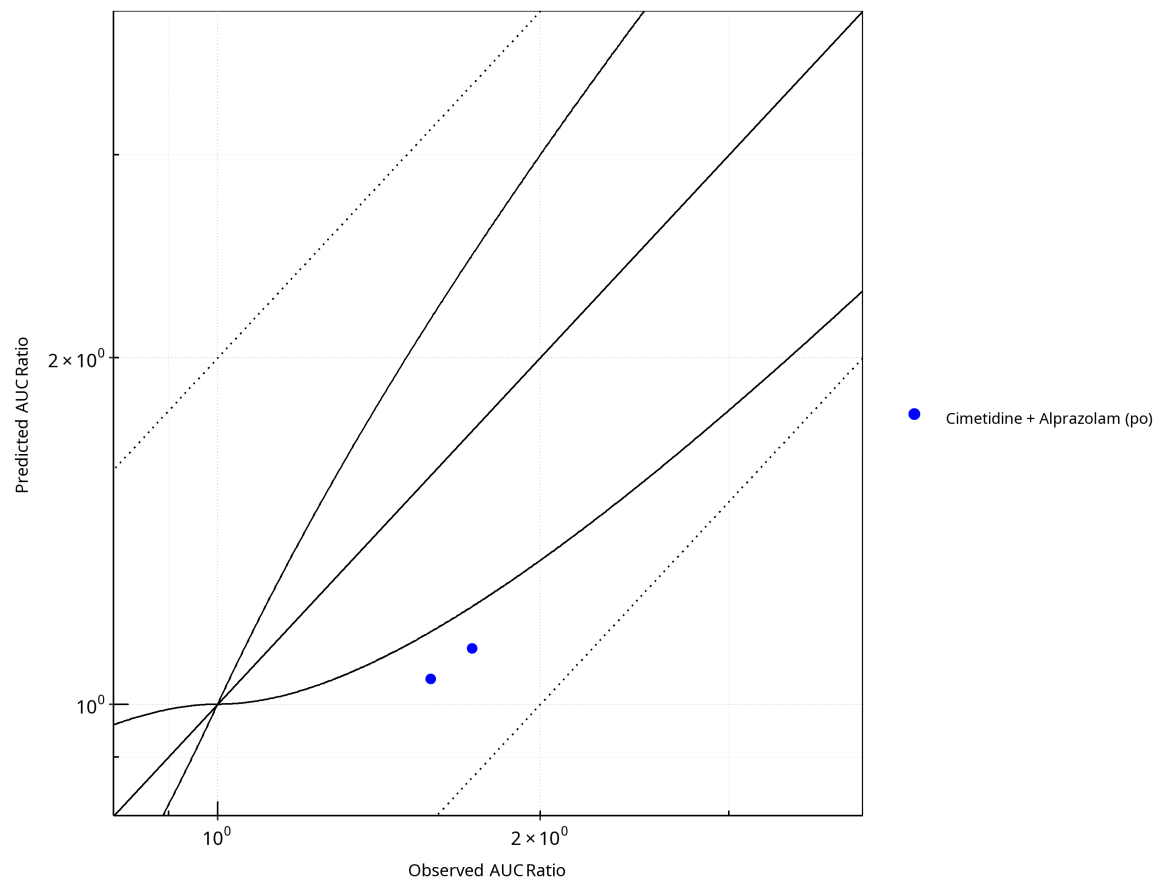


Figure 1-1: Plot. Predicted vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

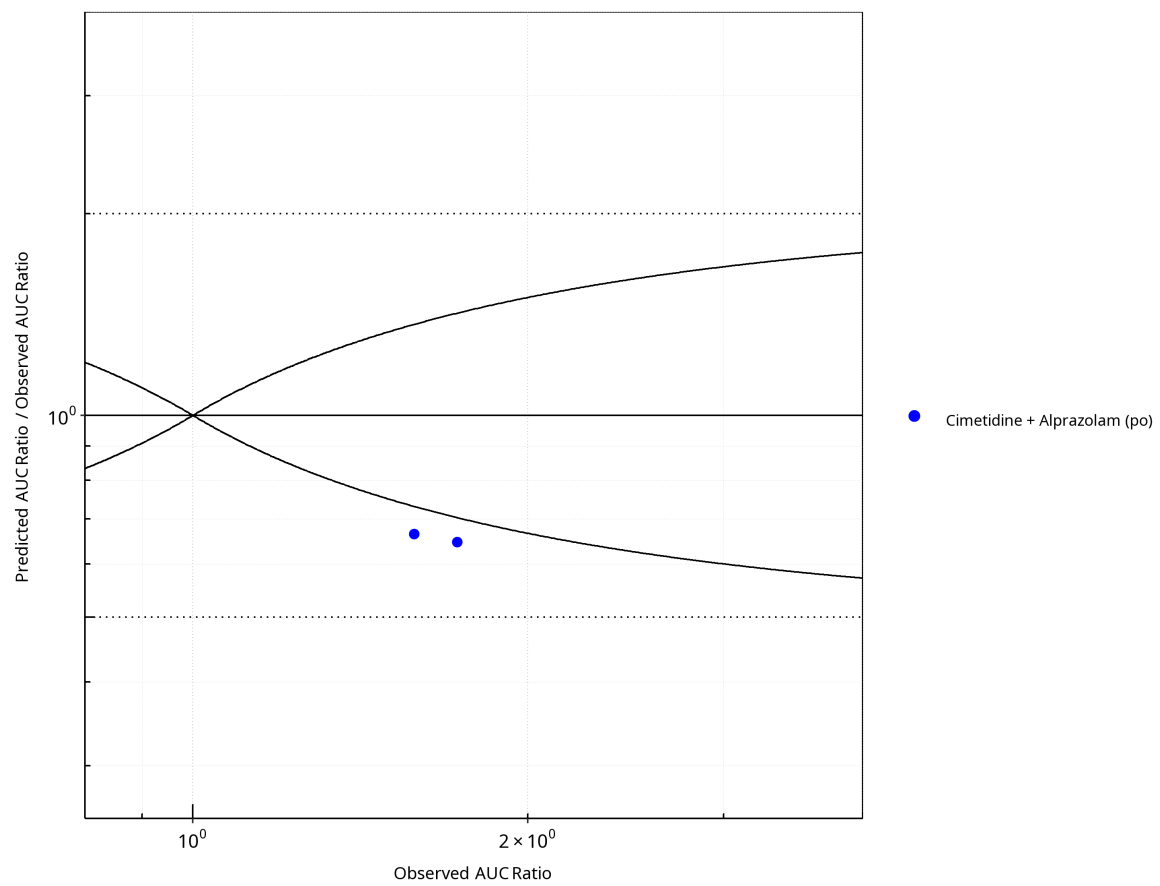


Figure 1-2: Plot. Predicted/Observed vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

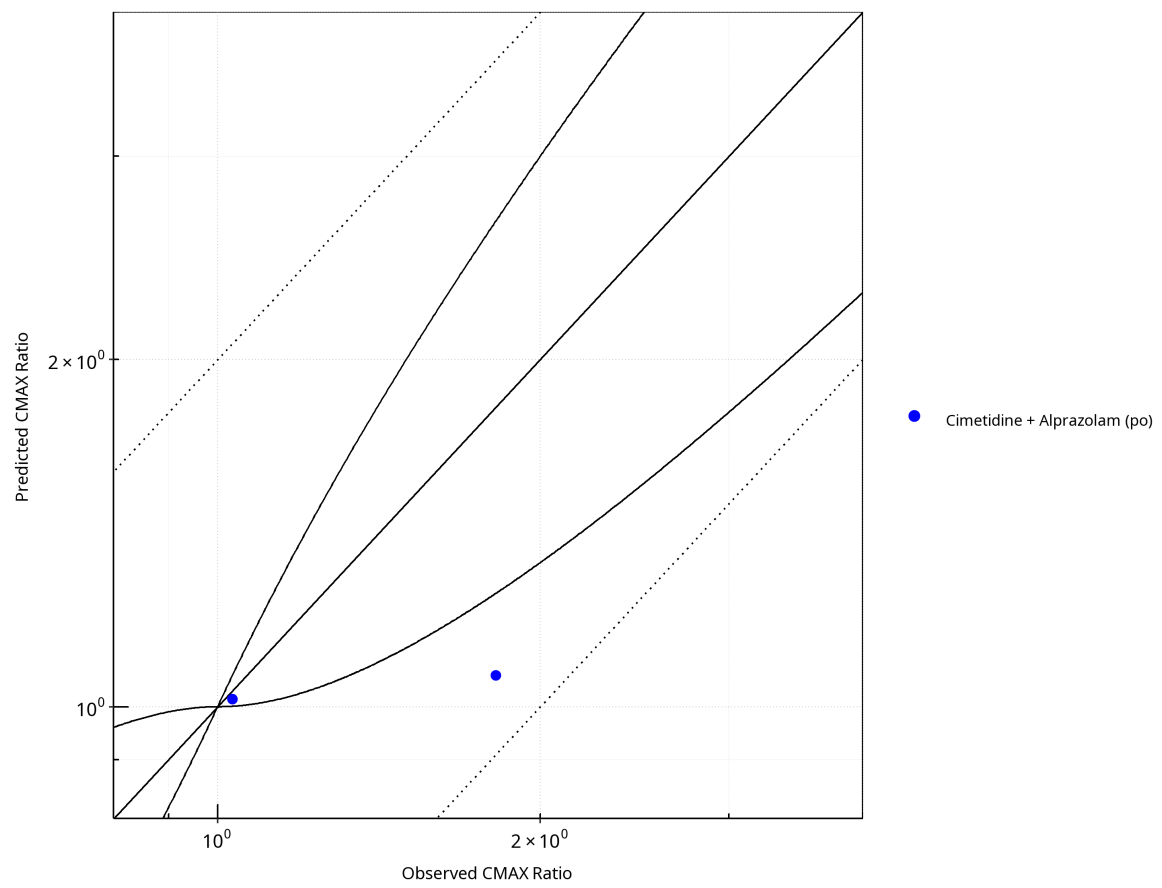


Figure 1-3: Plot. Predicted vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

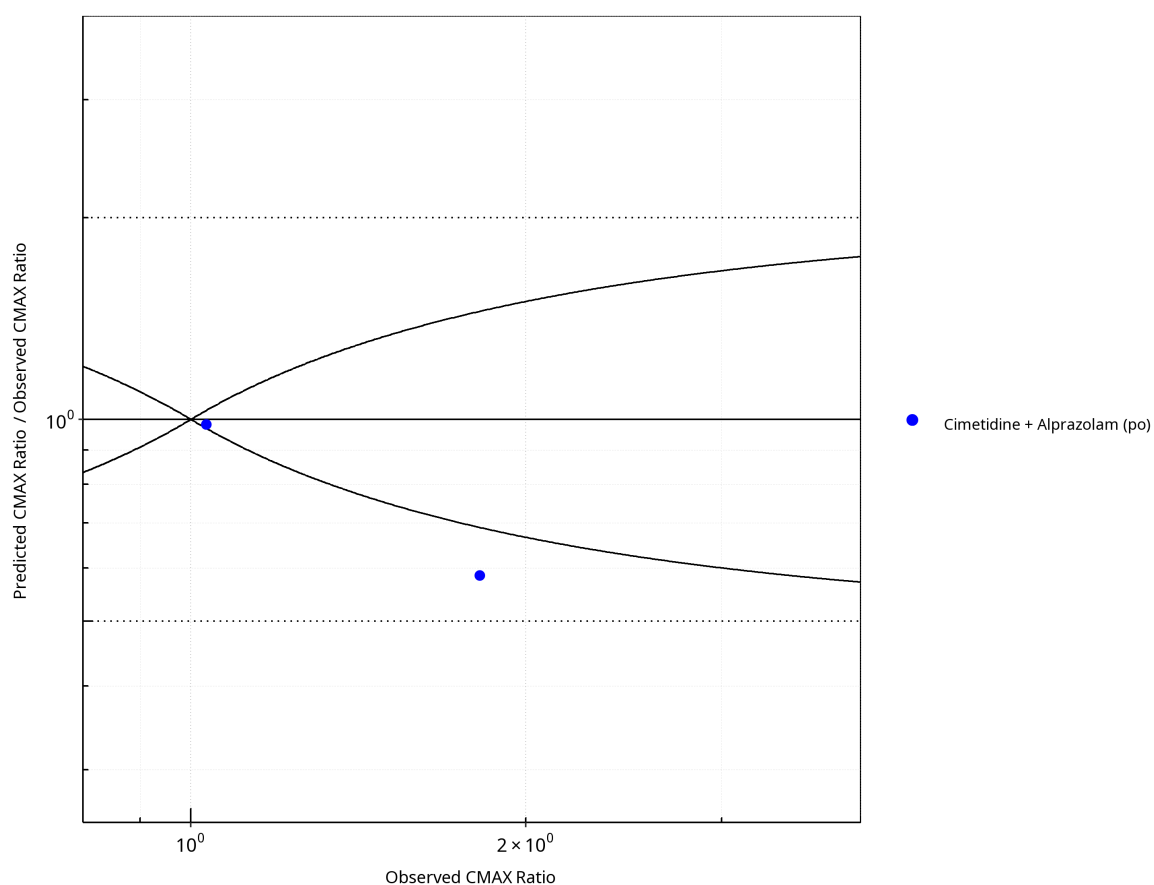


Figure 1-4: Plot. Predicted/Observed vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

Table 1-1: GMFE for Plot Ratio

PK parameter	GMFE
AUC	1.52
CMAX	1.32

Table 1-2: Summary table for Plot - AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

AUC	Number	Ratio [%]
Points total	2	-
Points within Guest <i>et al.</i>	0	0
Points within 2 fold	2	100

Table 1-3: Summary table for Plot - CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

CMAX	Number	Ratio [%]
Points total	2	-
Points within Guest <i>et al.</i>	1	50
Points within 2 fold	2	100

Table 1-4: Summary table for Plot

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted CMAX Ratio	Observed CMAX Ratio	Pred/Obs CMAX Ratio	Reference
1332	Cimetidine, 300 mg, PO, MD QID (1 day)	Alprazolam, PO	1.05	1.58	0.67	1.02	1.03	0.98	Abernethy 1983
1340	Cimetidine, 200/400 mg, PO, (200mg): MD TID (17 days); (400mg): OD (17 days)	Alprazolam, PO	1.12	1.73	0.65	1.07	1.82	0.59	Pourbaix 1985

1.1 Mechanism

1.1.1 Reversible Inhibition

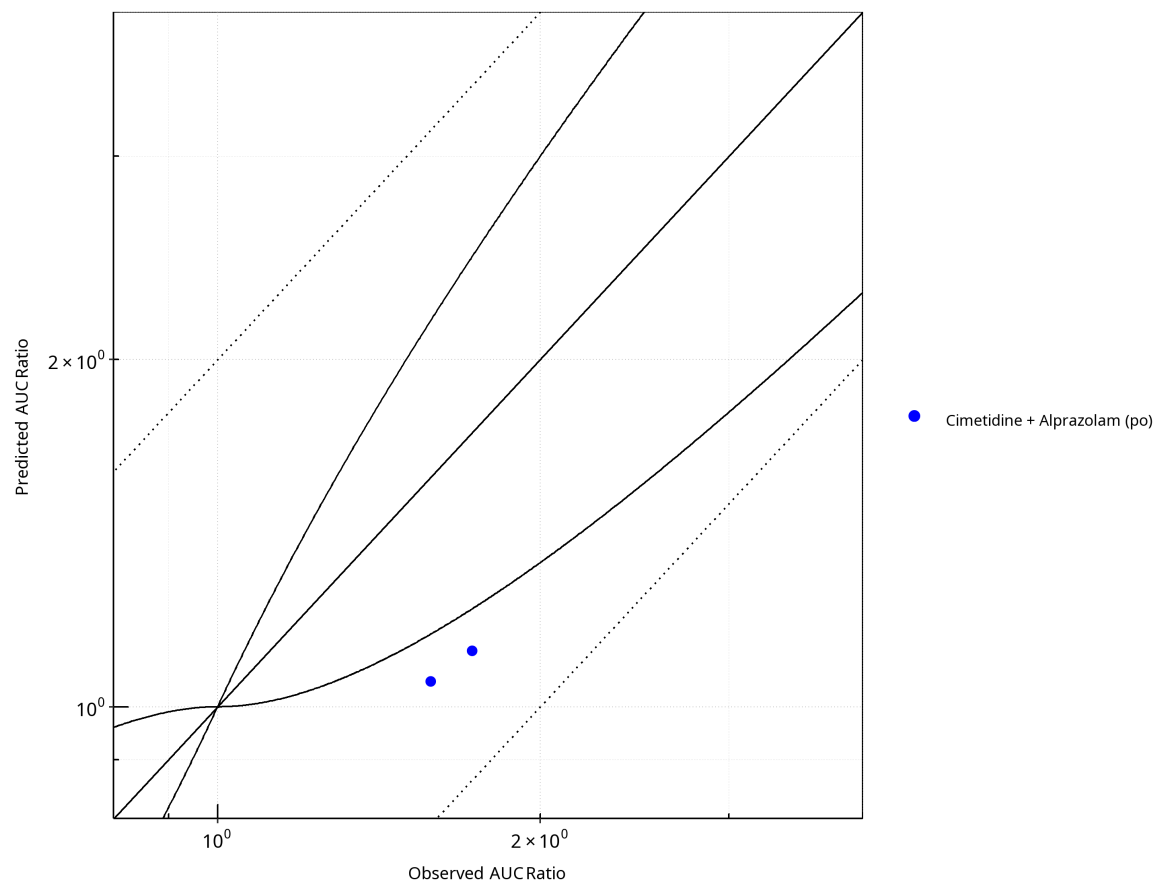


Figure 1-5: Plot. Mechanism: Reversible Inhibition. Predicted vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

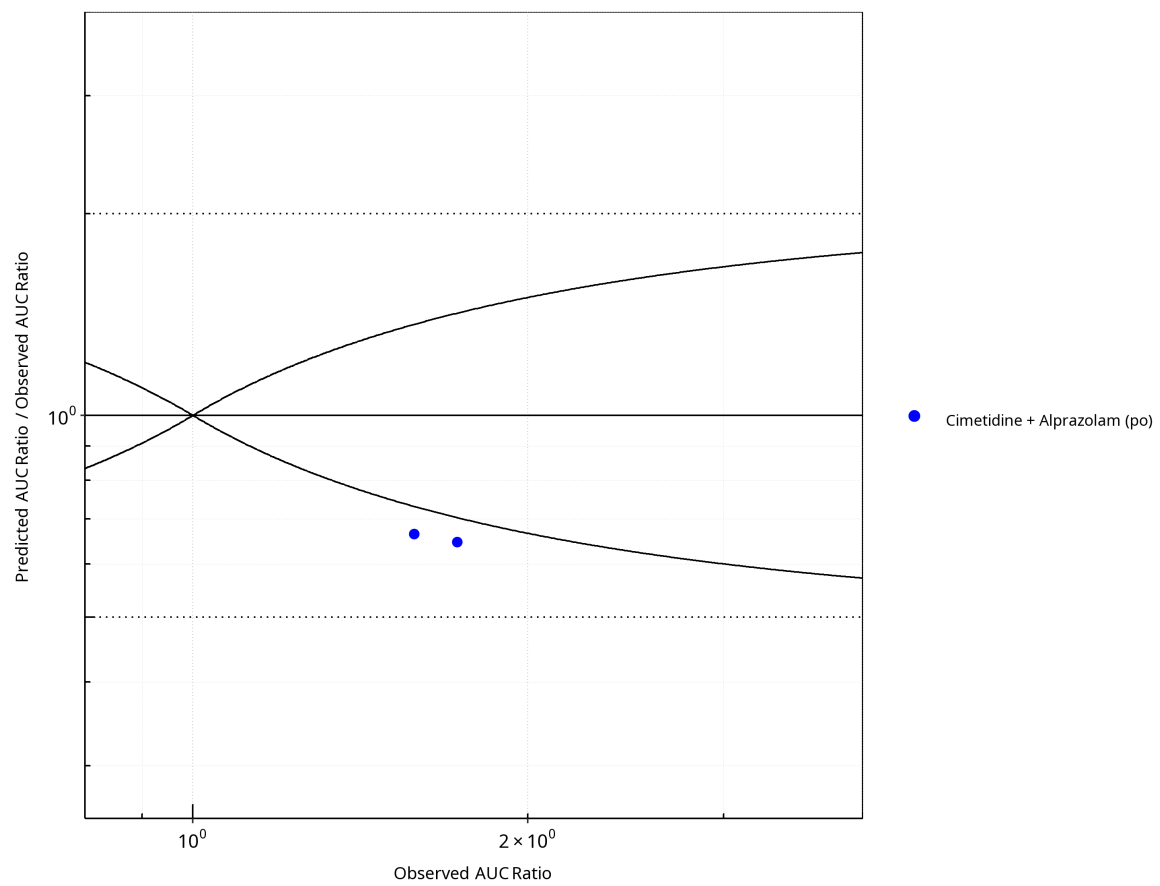


Figure 1-6: Plot. Mechanism: Reversible Inhibition. Predicted/Observed vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

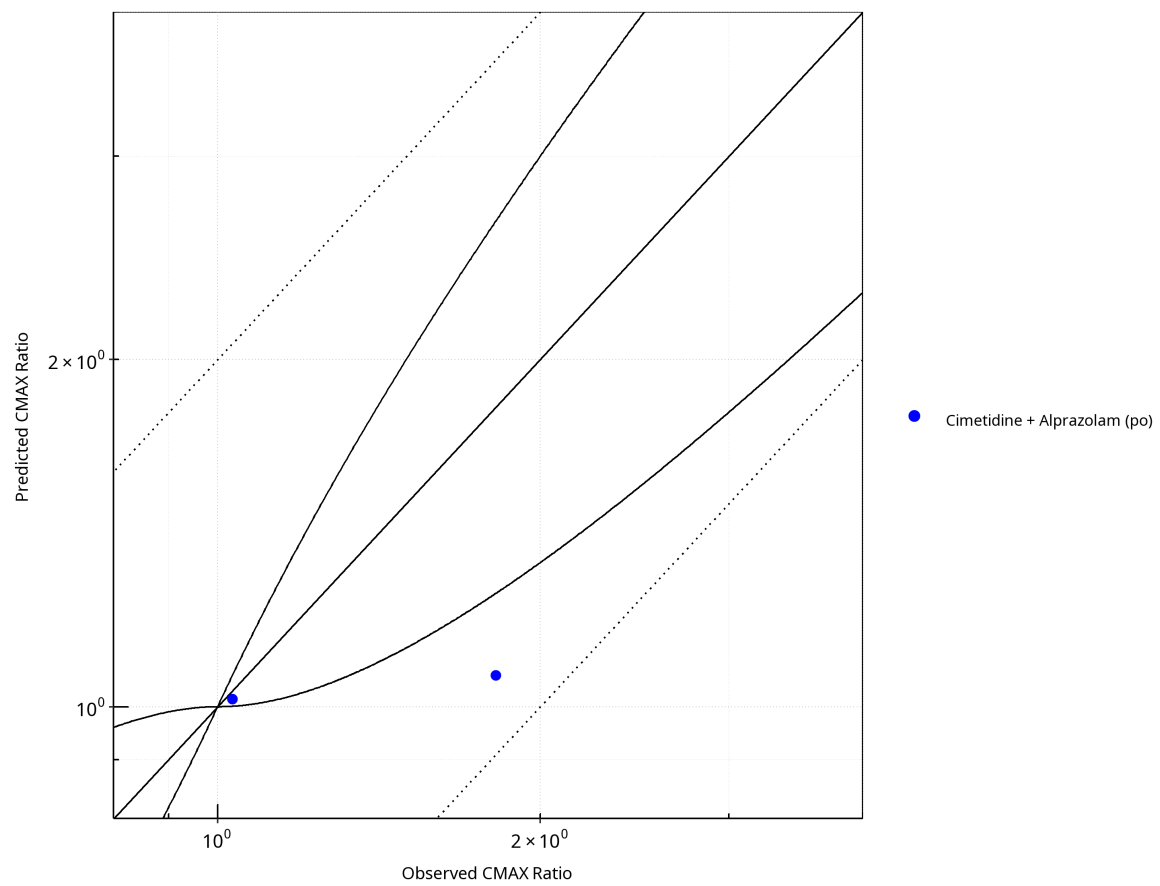


Figure 1-7: Plot. Mechanism: Reversible Inhibition. Predicted vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

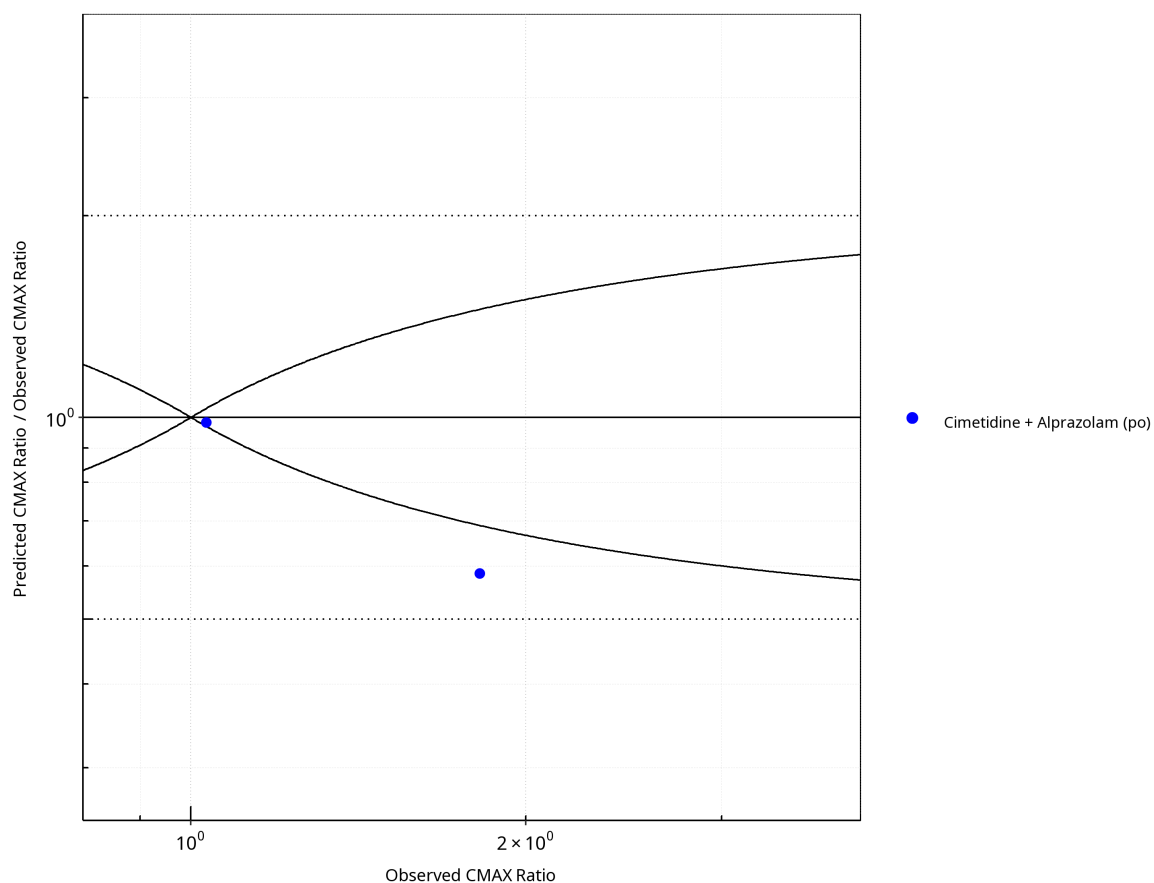


Figure 1-8: Plot. Mechanism: Reversible Inhibition. Predicted/Observed vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

Table 1-5: GMFE for Plot Ratio

PK parameter	GMFE
AUC	1.52
CMAX	1.32

Table 1-6: Summary table for Plot - AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

AUC	Number	Ratio [%]
Points total	2	-
Points within Guest <i>et al.</i>	0	0
Points within 2 fold	2	100

Table 1-7: Summary table for Plot - CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

CMAX	Number	Ratio [%]
Points total	2	-
Points within Guest <i>et al.</i>	1	50
Points within 2 fold	2	100

1.2 Perpetrator

1.2.1 Cimetidine

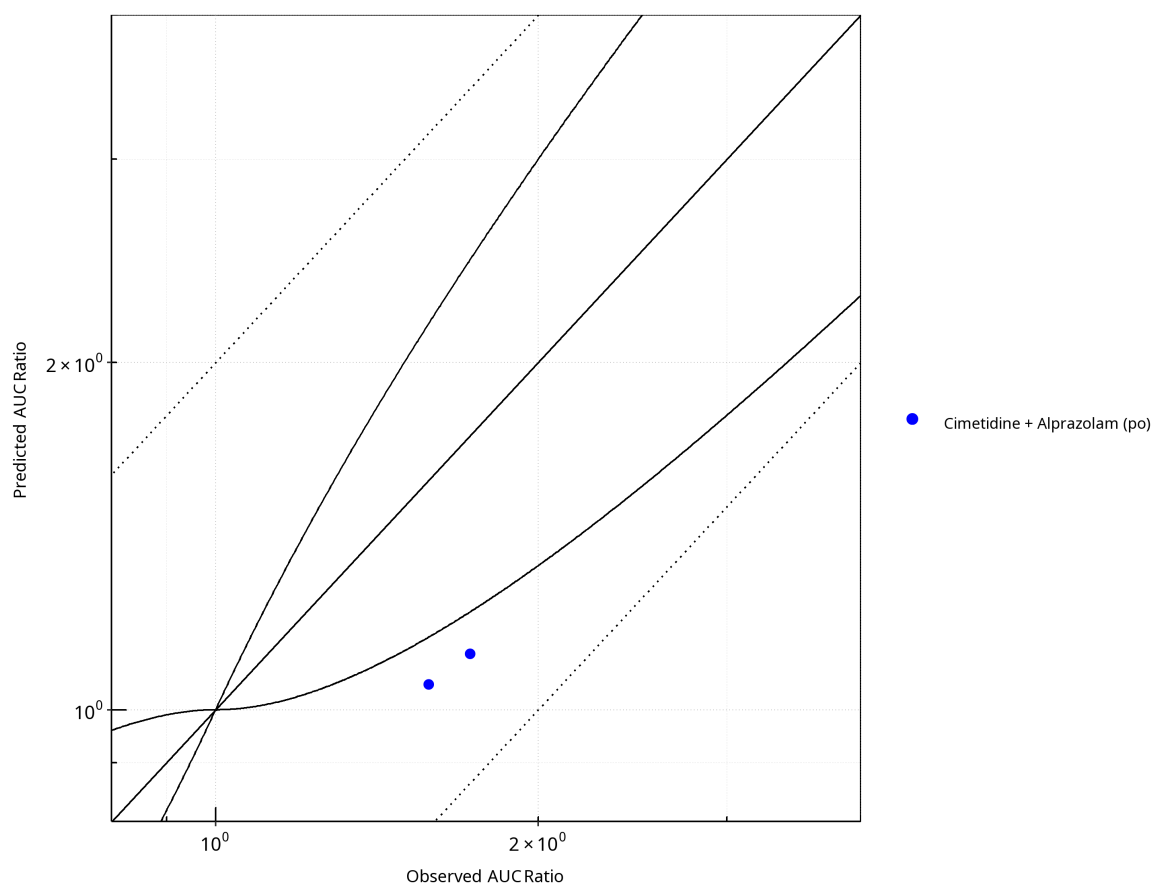


Figure 1-9: Plot. Perpetrator: Cimetidine. Predicted vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

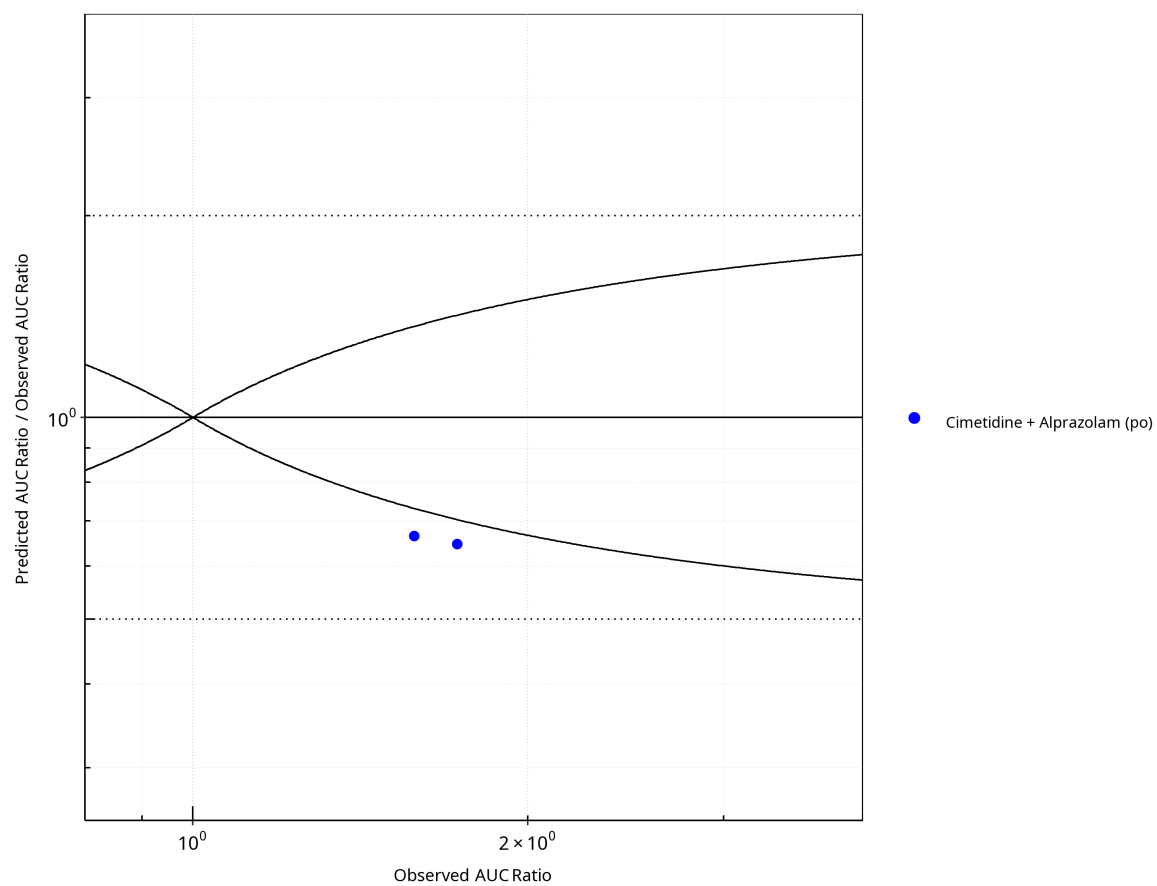


Figure 1-10: Plot. Perpetrator: Cimetidine. Predicted/Observed vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

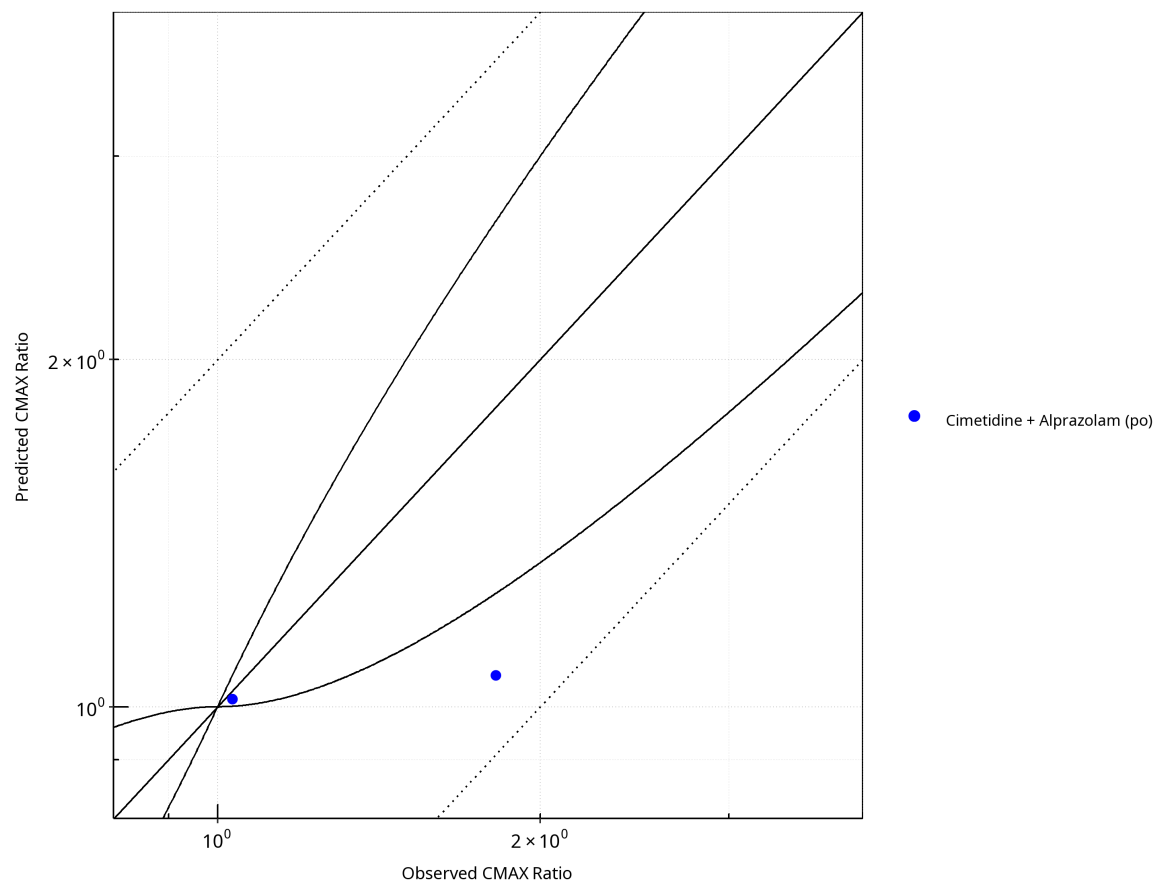


Figure 1-11: Plot. Perpetrator: Cimetidine. Predicted vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

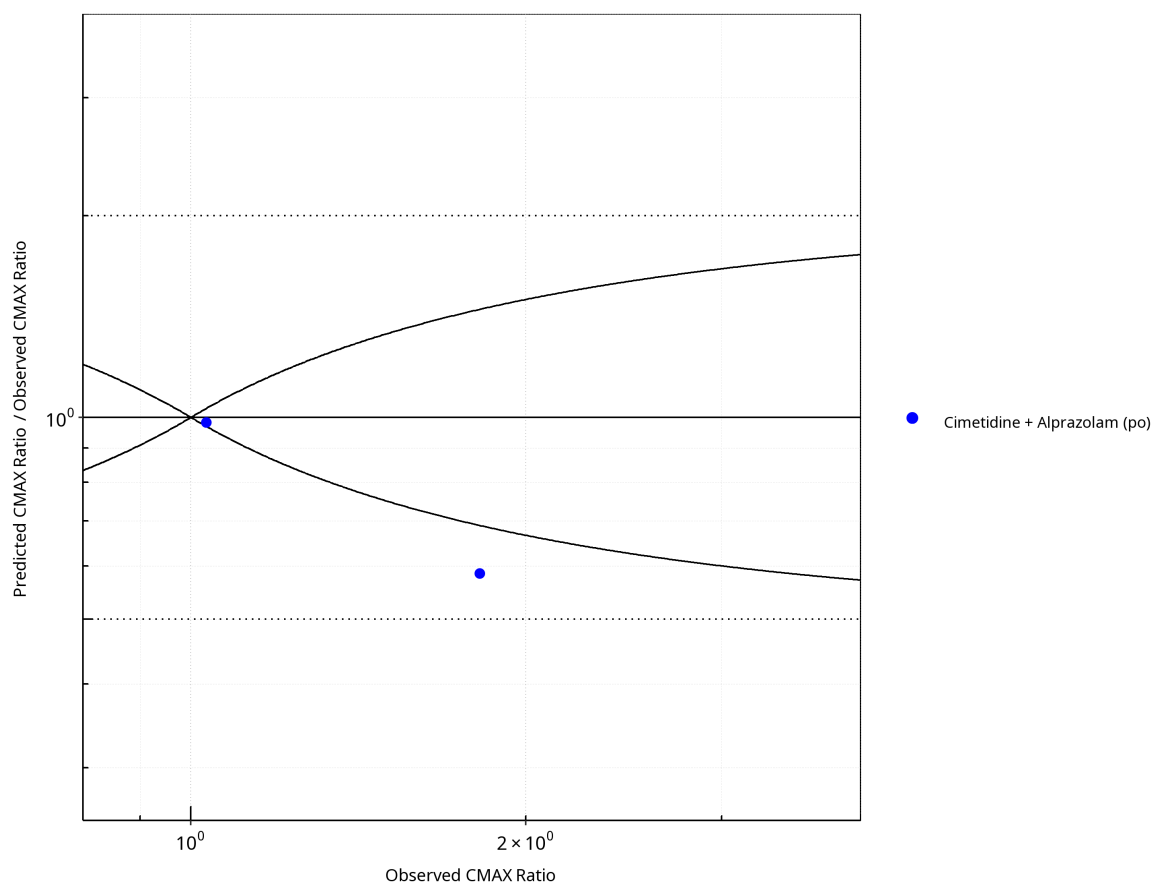


Figure 1-12: Plot. Perpetrator: Cimetidine. Predicted/Observed vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

Table 1-8: GMFE for Plot Ratio

PK parameter	GMFE
AUC	1.52
CMAX	1.32

Table 1-9: Summary table for Plot - AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

AUC	Number	Ratio [%]
Points total	2	-
Points within Guest <i>et al.</i>	0	0
Points within 2 fold	2	100

Table 1-10: Summary table for Plot - CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

CMAX	Number	Ratio [%]
Points total	2	-
Points within Guest <i>et al.</i>	1	50
Points within 2 fold	2	100

1.3 Victim

1.3.1 Alprazolam

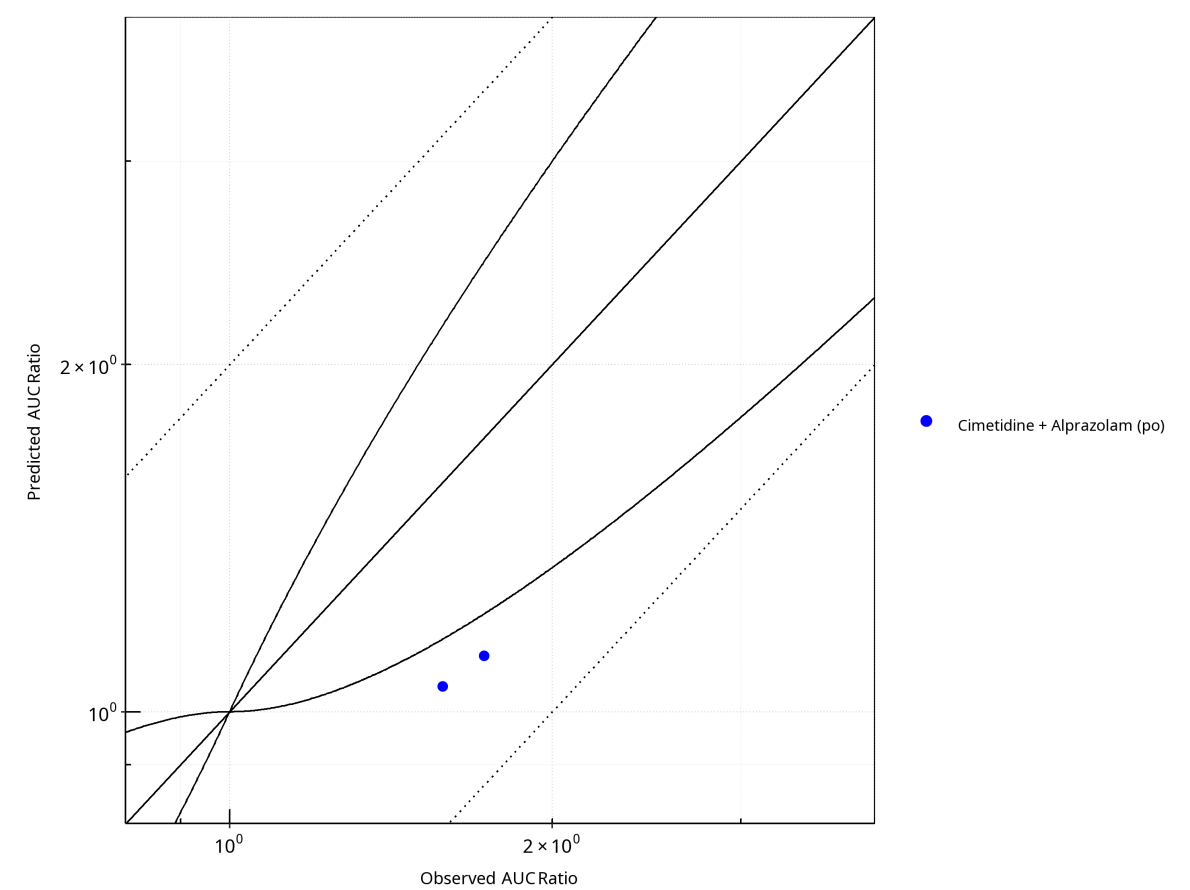


Figure 1-13: Plot. Victim: Alprazolam. Predicted vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

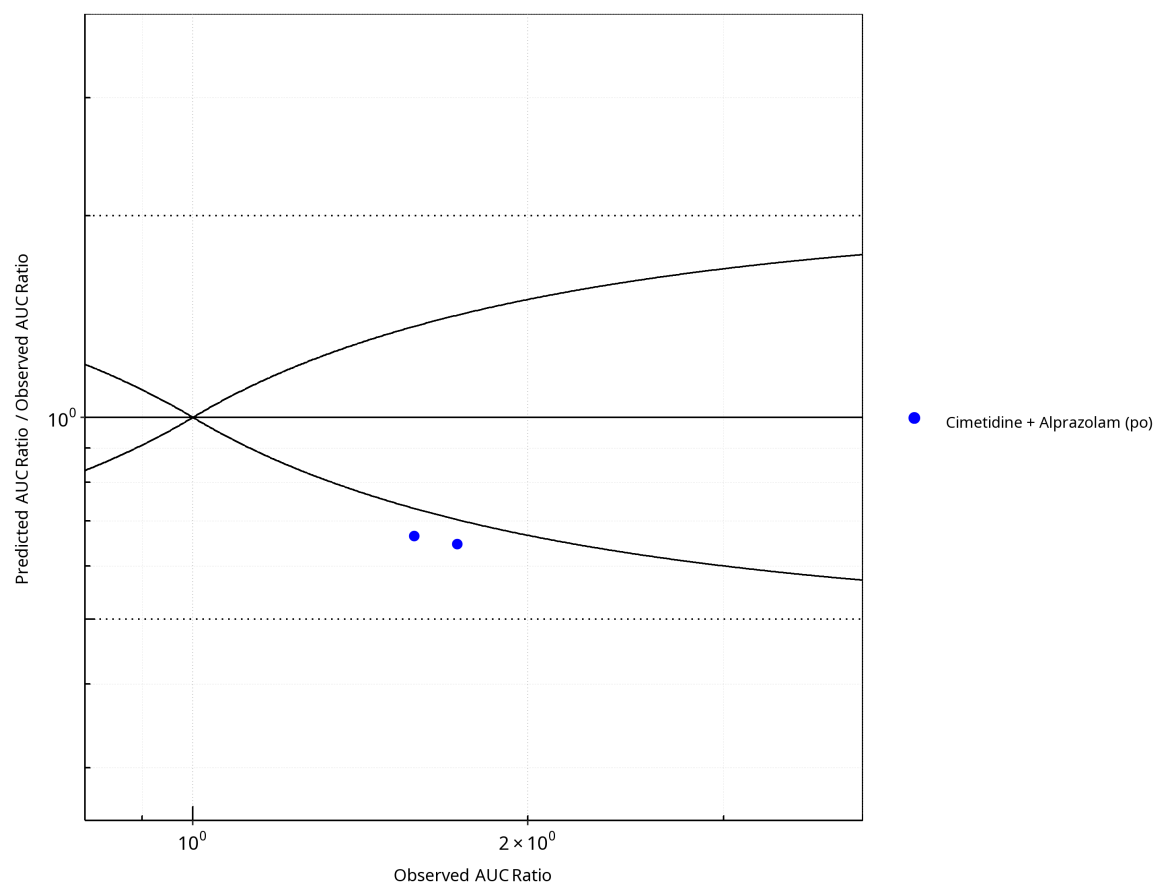


Figure 1-14: Plot. Victim: Alprazolam. Predicted/Observed vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

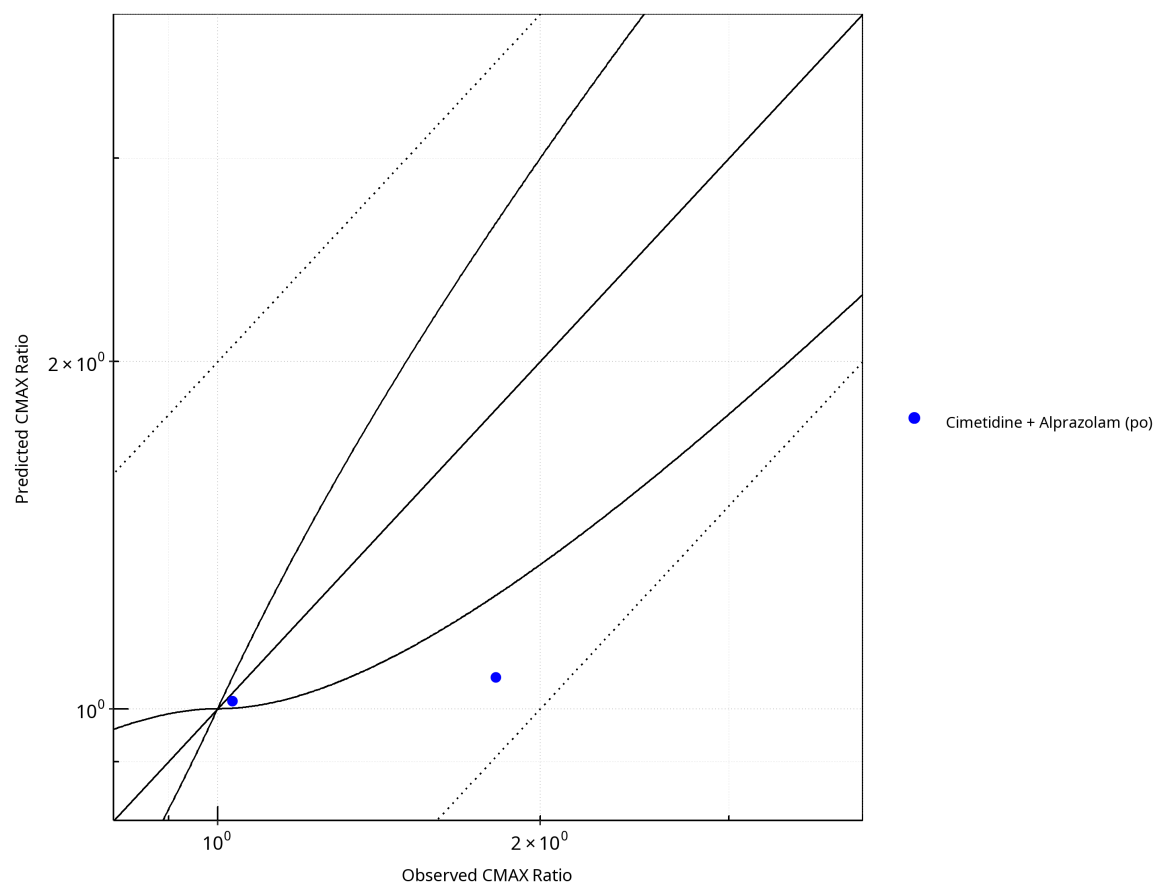


Figure 1-15: Plot. Victim: Alprazolam. Predicted vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

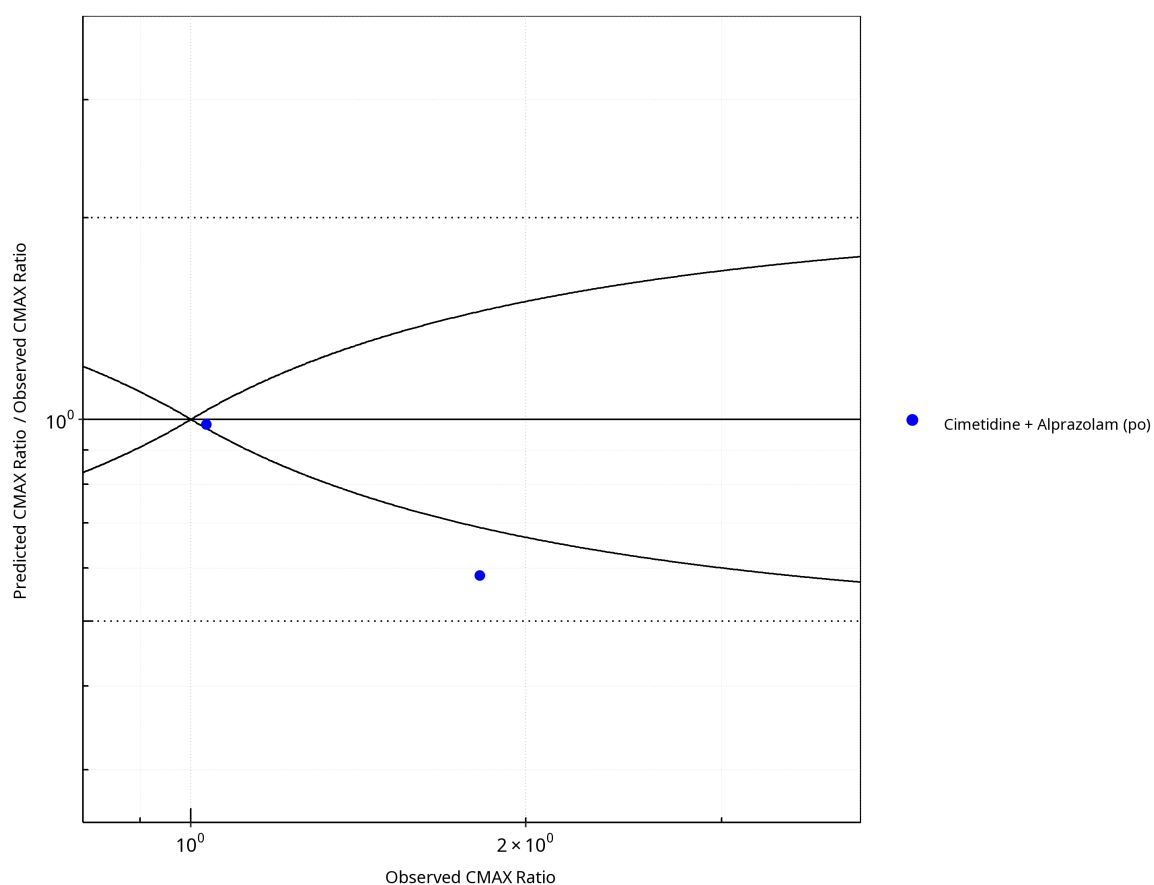


Figure 1-16: Plot. Victim: Alprazolam. Predicted/Observed vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

Table 1-11: GMFE for Plot Ratio

PK parameter	GMFE
AUC	1.52
CMAX	1.32

Table 1-12: Summary table for Plot - AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

AUC	Number	Ratio [%]
Points total	2	-
Points within Guest <i>et al.</i>	0	0
Points within 2 fold	2	100

Table 1-13: Summary table for Plot - CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

CMAX	Number	Ratio [%]
Points total	2	-
Points within Guest <i>et al.</i>	1	50
Points within 2 fold	2	100